

# Homework 06 - Labeling legislative bills

INFO 4940/5940 - Fall 2025

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## Exercise 1

```
# scripts/leg-label.py

import pandas as pd
from dotenv import load_dotenv
from pydantic import BaseModel
import pyarrow.feather as feather
import chatlas as ctl
from chatlas import batch_chat_structured

load_dotenv()

INPUT_FEATHER = "data/leg_lite.feather"
OUTPUT_CSV = "data/llm_predictions.csv"

PROMPT_SIMPLE = "prompts/cap-simple.md"
PROMPT_DETAILED = "prompts/cap-detailed.md"
PROMPT_REASONING = "prompts/cap-reasoning.md"

# Testing controls
TEST_ONLY = False
TEST_N = 3
MODELS = {
    "gpt-4.1": "gpt-4.1",
    "gpt-5-nano": "gpt-5-nano",
    "gpt-5": "gpt-5",
}

PROMPTS = {
    "naive": PROMPT_SIMPLE,
```

```

    "detailed": PROMPT_DETAILED,
    "reasoning": PROMPT_REASONING,
}

def load_prompt(path: str) -> str:
    """Load prompt from a markdown/text file."""
    with open(path, encoding="utf-8") as f:
        return f.read()

# Structured output

class PolicyPrediction(BaseModel):
    """Structured output for CAP classification."""
    policy_number: int
    reasoning: str | None = None

def main():
    # Load Data
    print(f"Loading data from {INPUT_FEATHER} ...")
    df_full = feather.read_feather(INPUT_FEATHER)

    expected_cols = {"description", "policy"}
    missing = expected_cols - set(df_full.columns)
    if missing:
        raise ValueError(f"Missing expected column(s) in leg_lite.feather: {missing}")

    df_full = df_full.reset_index(drop=True).copy()
    df_full["row_id"] = df_full.index
    df_full["policy"] = df_full["policy"].astype(int)

    if TEST_ONLY:
        df = df_full.iloc[:TEST_N].copy()
        print(f"TEST_ONLY is True → running on first {len(df)} rows")
    else:
        df = df_full
        print(f"TEST_ONLY is False → running on all {len(df)} rows")

    all_results: list[dict] = []

    # Loop over model × prompt, use batch_chat_structured
    for model_label, model_id in MODELS.items():
        for prompt_label, prompt_path in PROMPTS.items():
            print(f"\n=== Running batch for model={model_label} prompt={prompt_label} ===")

```

```

system_prompt = load_prompt(prompt_path)

chat = ctl.ChatOpenAI(
    model=model_id,
    system_prompt=system_prompt,
)

#Descriptions
prompts = df["description"].tolist()

# Path to store
batch_state_path = f"data/batch-{model_label}-{prompt_label}.json"

print(f"Submitting batch job → {batch_state_path}")

predictions = batch_chat_structured(
    chat=chat,
    prompts=prompts,
    path=batch_state_path,
    data_model=PolicyPrediction,
    wait=True,
)

if len(predictions) != len(df):
    raise RuntimeError(
        f"Expected {len(df)} predictions, got {len(predictions)}"
    )

# Attach predictions back to rows
for row, pred in zip(df.itertuples(index=False), predictions):
    if pred is None:
        pred_label = -1
        reasoning = "Batch request failed for this row."
    else:
        pred_label = int(pred.policy_number)
        reasoning = pred.reasoning or ""

    all_results.append(
        {
            "row_id": int(row.row_id),
            "description": row.description,
            "true_label": int(row.policy),
            "model_label": model_label,
            "model_id": model_id,
            "prompt_label": prompt_label,

```

```

        "pred_label": pred_label,
        "reasoning": reasoning,

        "total_tokens": None,
        "input_tokens": None,
        "output_tokens": None,
    }

)

# save as csv
results_df = pd.DataFrame(all_results)
print(f"\nSaving predictions to {OUTPUT_CSV} ...")
results_df.to_csv(OUTPUT_CSV, index=False)
print("Done.")

if __name__ == "__main__":
    mode = "TEST" if TEST_ONLY else "FULL"
    print(f"Running scripts/leg-label.py (chatlas batch_chat_structured,
mode={mode})")
    main()

```

## Naive and simple prompt

You classify U.S. congressional bill descriptions into one of the 20 major policy topics used by the Comparative Agendas Project.

Read the description and decide which single policy topic number (1–20) best matches the bill’s main purpose. Focus on the primary policy area, not minor details or boilerplate phrases like “and for other purposes.” If the text touches multiple issues, choose the topic that is most central to the bill’s intent.

## Explicit code/label values

You are an expert policy coder for the Comparative Agendas Project (CAP). Your job is to assign each U.S. congressional bill description to exactly one CAP major topic based on its primary policy focus.

Use the bill’s main purpose to choose the best-fitting topic number. Ignore boilerplate language like “and for other purposes” and secondary side effects. When multiple topics appear, pick the one that is most central to what the bill is trying to accomplish.

Here are the CAP major policy topic codes:

- 1 – Macroeconomics
- 2 – Civil rights, minority issues, civil liberties
- 3 – Health
- 4 – Agriculture
- 5 – Labor and employment

- 6 – Education
- 7 – Environment
- 8 – Energy
- 9 – Immigration
- 10 – Transportation
- 11 – Law, crime, family issues
- 12 – Social welfare
- 13 – Community development and housing
- 14 – Banking, finance, domestic commerce
- 15 – Defense
- 16 – Space, technology, communications
- 17 – Foreign trade
- 18 – International affairs, foreign aid
- 19 – Government operations
- 20 – Public lands and water management

### **Detailed and reasoning prompt**

You are an expert policy analyst trained in the Comparative Agendas Project (CAP) coding rules. Your task is to assign a U.S. congressional bill description to one CAP major policy topic (1–20) based on the bill’s primary purpose.

Follow this process when you classify:

1. Read the description carefully and identify the core policy problem the bill addresses.
2. Match that problem to the closest CAP major topic, focusing on the main goal rather than minor side effects or boilerplate phrases like “and for other purposes.”
3. If several topics are mentioned, choose the single topic that best captures the central intent of the bill.
4. Be prepared to briefly explain why this topic is more appropriate than nearby alternatives (for example, energy vs. environment, health vs. social welfare, etc.).

Here are the CAP major policy topic codes:

- 1 – Macroeconomics
- 2 – Civil rights, minority issues, civil liberties
- 3 – Health
- 4 – Agriculture
- 5 – Labor and employment
- 6 – Education
- 7 – Environment
- 8 – Energy
- 9 – Immigration
- 10 – Transportation
- 11 – Law, crime, family issues
- 12 – Social welfare

- 13 – Community development and housing
- 14 – Banking, finance, domestic commerce
- 15 – Defense
- 16 – Space, technology, communications
- 17 – Foreign trade
- 18 – International affairs, foreign aid
- 19 – Government operations
- 20 – Public lands and water management

## Exercise 2

```
import pyarrow.feather as feather

import numpy as np
from sklearn.metrics import accuracy_score, f1_score, confusion_matrix

import pandas as pd

preds = pd.read_csv("data/llm_predictions.csv")

# Make sure types are right
preds["true_label"] = preds["true_label"].astype(int)
preds["pred_label"] = preds["pred_label"].astype(int)

def multiclass_sensitivity_specificity(y_true, y_pred):
    """
    Compute macro-averaged sensitivity (recall) and specificity
    for a multiclass problem using the confusion matrix.
    """
    labels = np.unique(y_true)
    cm = confusion_matrix(y_true, y_pred, labels=labels)

    sensitivities = []
    specificities = []

    for i, _ in enumerate(labels):
        tp = cm[i, i]
        fn = cm[i, :].sum() - tp
        fp = cm[:, i].sum() - tp
        tn = cm.sum() - tp - fn - fp

        sens = tp / (tp + fn) if (tp + fn) > 0 else np.nan
        spec = tn / (tn + fp) if (tn + fp) > 0 else np.nan

        sensitivities.append(sens)
```

```

        specificities.append(spec)

    return float(np.nanmean(sensitivities)), float(np.nanmean(specificities))

rows = []

for (model, prompt), group in preds.groupby(["model_label", "prompt_label"]):
    y_true = group["true_label"].to_numpy()
    y_pred = group["pred_label"].to_numpy()

    acc = accuracy_score(y_true, y_pred)
    f1 = f1_score(y_true, y_pred, average="macro")
    sens, spec = multiclass_sensitivity_specificity(y_true, y_pred)

    rows.append(
        {
            "model": model,
            "prompt": prompt,
            "accuracy": acc,
            "f1_macro": f1,
            "sensitivity_macro": sens,
            "specificity_macro": spec,
        }
    )

metrics_df = pd.DataFrame(rows).sort_values(["model",
"prompt"]).reset_index(drop=True)
metrics_df

```

|   | model      | prompt    | accuracy | f1_macro | sensitivity_macro | specificity_macro |
|---|------------|-----------|----------|----------|-------------------|-------------------|
| 0 | gpt-4.1    | detailed  | 0.648    | 0.624376 | 0.619057          | 0.981182          |
| 1 | gpt-4.1    | naive     | 0.222    | 0.183037 | 0.197199          | 0.958887          |
| 2 | gpt-4.1    | reasoning | 0.640    | 0.612605 | 0.594063          | 0.980736          |
| 3 | gpt-5      | detailed  | 0.716    | 0.667050 | 0.664353          | 0.984804          |
| 4 | gpt-5      | naive     | 0.640    | 0.593111 | 0.604821          | 0.980917          |
| 5 | gpt-5      | reasoning | 0.728    | 0.670488 | 0.669870          | 0.985477          |
| 6 | gpt-5-nano | detailed  | 0.318    | 0.345126 | 0.580041          | 0.979487          |
| 7 | gpt-5-nano | naive     | 0.050    | 0.046698 | 0.062564          | 0.950094          |
| 8 | gpt-5-nano | reasoning | 0.658    | 0.596047 | 0.638995          | 0.982055          |

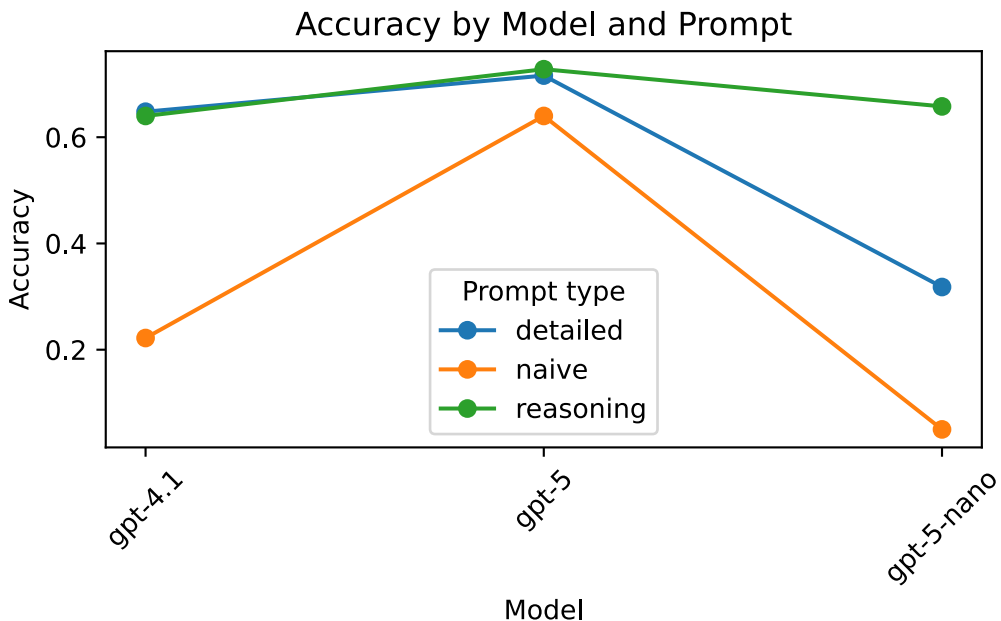
```
import matplotlib.pyplot as plt
```

```

plt.figure()
for prompt in metrics_df["prompt"].unique():
    subset = metrics_df[metrics_df["prompt"] == prompt]
    plt.plot(subset["model"], subset["accuracy"], marker="o", label=prompt)

plt.ylabel("Accuracy")
plt.xlabel("Model")
plt.title("Accuracy by Model and Prompt")
plt.legend(title="Prompt type")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

```



```

import matplotlib.pyplot as plt
import numpy as np

# Make sure metrics_df exists from previous chunk

models = metrics_df["model"].unique()
prompts = metrics_df["prompt"].unique()

x = np.arange(len(models)) # positions for models on x-axis
width = 0.25               # width of each bar

plt.figure()

```

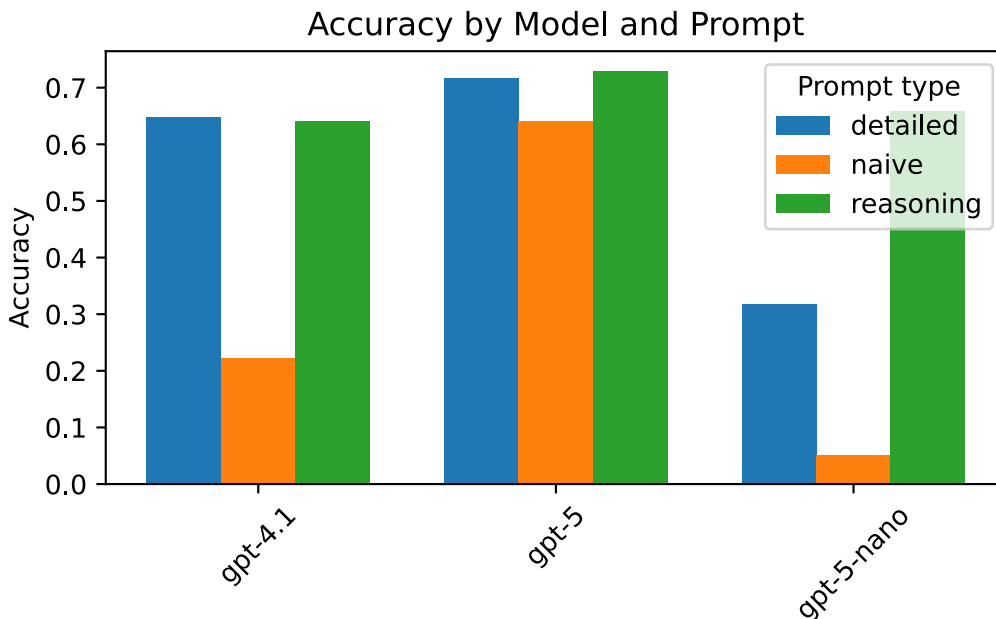


```

for i, prompt in enumerate(prompts):
    subset = metrics_df[metrics_df["prompt"] == prompt]
    # Align subset to models order
    subset = subset.set_index("model").loc[models]
    plt.bar(x + i * width, subset["accuracy"], width=width, label=prompt)

plt.xticks(x + width, models, rotation=45)
plt.ylabel("Accuracy")
plt.title("Accuracy by Model and Prompt")
plt.legend(title="Prompt type")
plt.tight_layout()
plt.show()

```



```

import matplotlib.pyplot as plt

# Pivot F1 into a matrix: rows = prompt, cols = model

f1_pivot = metrics_df.pivot(index="prompt", columns="model",
                             values="f1_macro")

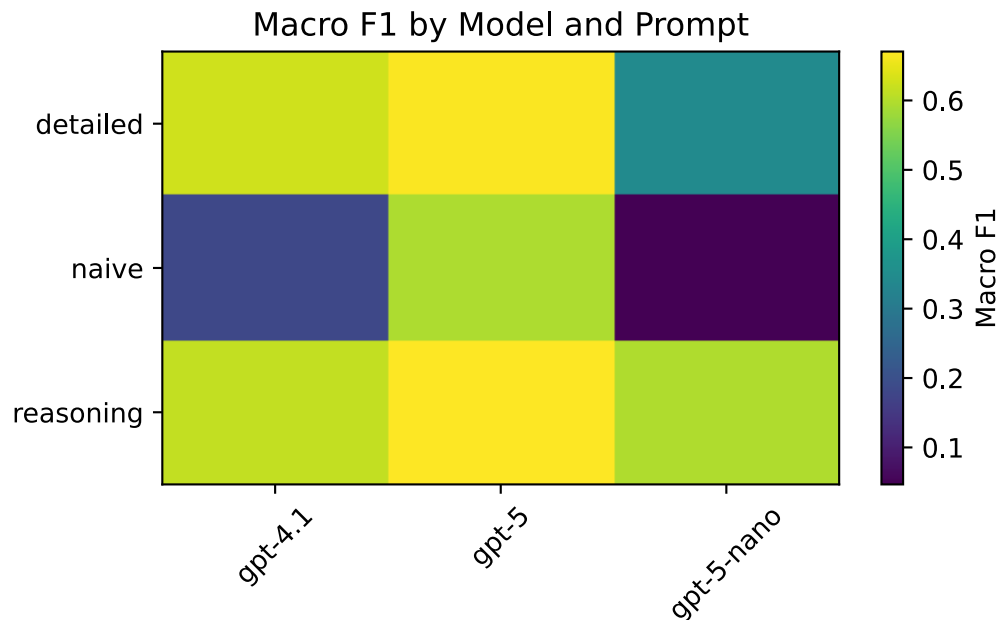
plt.figure()
plt.imshow(f1_pivot.values, aspect="auto")
plt.colorbar(label="Macro F1")

plt.xticks(range(len(f1_pivot.columns)), f1_pivot.columns, rotation=45)

```

```
plt.yticks(range(len(f1_pivot.index)), f1_pivot.index)

plt.title("Macro F1 by Model and Prompt")
plt.tight_layout()
plt.show()
```



```
from sklearn.metrics import confusion_matrix

# Find the best combo by accuracy (you can change to f1_macro if you prefer)
best_row = metrics_df.sort_values("accuracy", ascending=False).iloc[0]
best_model = best_row["model"]
best_prompt = best_row["prompt"]
print("Best combo:", best_model, best_prompt)

# Filter predictions for that combo

best_preds = preds[
    (preds["model_label"] == best_model) & (preds["prompt_label"] ==
best_prompt)
]

y_true = best_preds["true_label"].astype(int).to_numpy()
y_pred = best_preds["pred_label"].astype(int).to_numpy()

labels = sorted(np.unique(y_true))
cm = confusion_matrix(y_true, y_pred, labels=labels)
```

```

plt.figure(figsize=(6, 6))
plt.imshow(cm, interpolation="nearest")
plt.title(f"Confusion Matrix for {best_model} / {best_prompt}")
plt.xlabel("Predicted label")
plt.ylabel("True label")
plt.xticks(range(len(labels)), labels, rotation=45)
plt.yticks(range(len(labels)), labels)

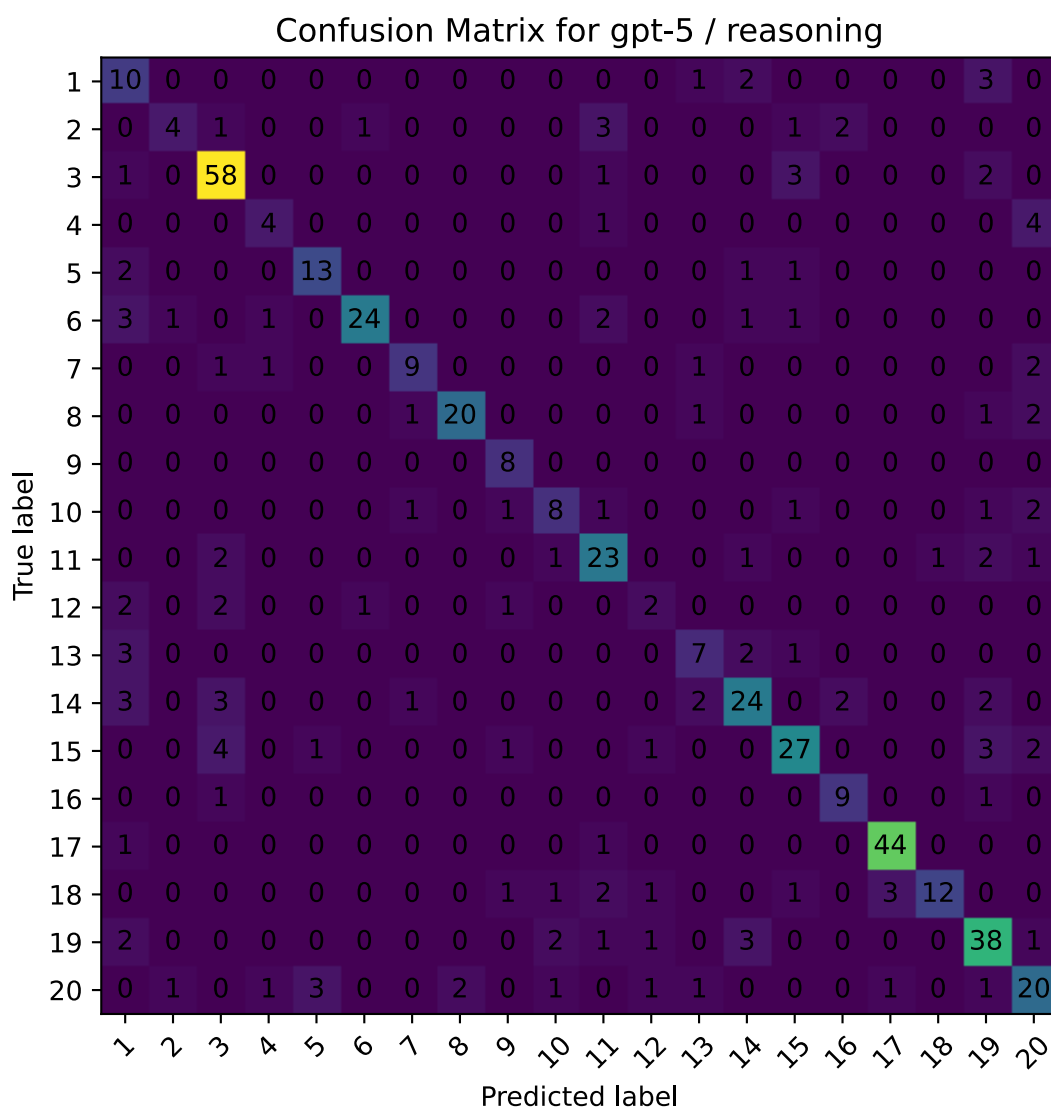
# Add counts on the heatmap

for i in range(len(labels)):
    for j in range(len(labels)):
        plt.text(j, i, cm[i, j], ha="center", va="center")

plt.tight_layout()
plt.show()

```

Best combo: gpt-5 reasoning



### Comparing Models

Overall, the reasoning and detailed prompts perform significantly better than naive prompts across all models, highlighting the importance of prompt engineering in leveraging LLM capabilities effectively. Across models, GPT-5 consistently outperforms GPT-4.1 and GPT-5-nano, indicating that larger models with more advanced architectures yield better classification performance. Sensitivity remains a challenge, particularly for smaller models and naive prompts, suggesting that these configurations struggle to identify positive instances effectively. Specificity is generally high across all models and prompts, indicating that negative instances are classified correctly more often. These insights underscore the value of using larger LLMs with well-crafted prompts for classification tasks to achieve optimal accuracy and balance between sensitivity and specificity.

I was unable to retain token usage with chatlas.

## Exercise 3

Link to app: <https://019a9d96-3e1e-232e-db76-84b5f6eba95f.share.connect.posit.cloud/>

```
library(shiny)
library(bslib)

ui <- page_fillable()

server <- function(input, output, session) {}

shinyApp(ui, server)
```

```
from __future__ import annotations

from textwrap import dedent

from shiny import App, ui, reactive
from dotenv import load_dotenv
from chatlas import ChatOpenAI

from rag import get_top_k_similar_documents

load_dotenv()

system_prompt = dedent("""
    You are a course tutor for INFO 4940 / 5940 at Cornell.

    Your role:
    - Answer questions about topics covered in this course (LLMs, prompts,
    RAG, evaluation, dashboards, etc.).
    - Help students reason through homework and projects WITHOUT giving full
    solutions.
    - Explain code and concepts clearly using R or Python examples when
    requested.
    - Use the provided context (syllabus, assignment text, project
    description) as your primary factual source.

    What you MUST do:
    - Be honest about what you know and what you don't.
    - Encourage good academic integrity: guide, don't solve.
    - Prefer course materials for questions about policies, grading, and due
    dates.
    - When giving code, keep it fairly short and explain it.

    What you MUST NOT do:
    - Do NOT write full homework or project answers that could be submitted
```

```

as-is.
    - Do NOT fabricate course policies or due dates.
    - Do NOT provide long, copy-pasteable solutions to graded questions.

Style:
    - Friendly, concise, and concrete.
    - If the student asks for a specific language, use that (R or Python).
    - If they don't specify, default to Python for code.
    - If their question is vague, ask one clarifying question before
    answering.
    """).strip()
#LLM

chat_client = ChatOpenAI(
    model="gpt-4.1-mini", # adjust to your allowed model
    system_prompt=system_prompt,
)

# UI

app_ui = ui.page_sidebar(
    ui.sidebar(
        ui.h4("Tutor settings"),

        ui.input_radio_buttons(
            "help_mode",
            "What do you need help with?",
            choices=[
                "Concepts (lectures/readings)",
                "Assignments & projects",
                "Code debugging",
                "Course policies (syllabus)",
                "Study strategies",
            ],
            selected="Concepts (lectures/readings)",
        ),

        ui.input_radio_buttons(
            "pref_lang",
            "Preferred language for examples",
            choices=["No preference", "Python", "R"],
            selected="No preference",
        ),

        ui.hr(),
        ui.input_action_button(
            "show_tips",
            "Show tips for good questions",

```

```

        class_="btn-secondary w-100",
    ),
    ui.br(),
    ui.input_action_button(
        "show_examples",
        "Show example questions",
        class_="btn-outline-secondary w-100",
    ),
),

ui.layout_columns(
    ui.card(
        ui.card_header("INFO 4940 / 5940 Tutor Bot"),
        ui.markdown(
            "Ask about lectures, homework, projects, "
            "course policies, or how to structure your code.\n\n"
        ),
        ui.chat_ui(
            "tutor_chat",
            placeholder="Ask your question here...",
        ),
    ),
),

title="INFO 4940 / 5940 Tutor Bot",
fillable=True,
fillable_mobile=True,
)

# Server

def server(input, output, session):

    chat = ui.Chat(id="tutor_chat")

    # Poupups

    tips_modal = ui.modal(
        ui.h4("Tips for using the tutor"),
        ui.tags.ul(
            ui.tags.li("Ask focused questions (e.g., one assignment or concept at a time)."),
            ui.tags.li("Mention the assignment number or topic when relevant."),
            ui.tags.li("If you're stuck on code, paste the error message and a small code snippet."),
        ),
        title="Tips",
    )

```

```

        easy_close=True,
        footer=ui.input_action_button(
            "close_tips",
            "Close",
            class_="btn-primary"
        ),
    )

    examples_modal = ui.modal(
        ui.h4("Example questions"),
        ui.tags.ul(
            ui.tags.li("What is RAG and how is it different from fine-
tuning?"),
            ui.tags.li("Can you remind me of the evaluation criteria for
Homework 4?"),
            ui.tags.li("How should I structure a system prompt for a labeling
task?"),
            ui.tags.li("What is the late policy for homework in this
class?"),
        ),
        title="Example questions",
        easy_close=True,
        footer=ui.input_action_button(
            "close_examples",
            "Close",
            class_="btn-primary"
        ),
    )

    @reactive.Effect
    @reactive.event(input.show_tips)
    def _show_tips():
        ui.modal_show(tips_modal)

    @reactive.Effect
    @reactive.event(input.close_tips)
    def _close_tips():
        ui.modal_remove()

    @reactive.Effect
    @reactive.event(input.show_examples)
    def _show_examples():
        ui.modal_show(examples_modal)

    @reactive.Effect
    @reactive.event(input.close_examples)
    def _close_examples():
        ui.modal_remove()

```



```

# Chat handler

@chat.on_user_submit
async def handle_user_input(user_input: str):
    help_mode = input.help_mode()
    pref_lang = input.pref_lang()

    # 1. Retrieve relevant course docs via RAG
    top_docs = get_top_k_similar_documents(user_input, top_k=4)

    if top_docs:
        context_block = "\n\n---\n\n".join(top_docs)
    else:
        context_block = "No course documents were retrieved."

    # 2. Build full prompt
    prompt = f"""You are tutoring an INFO 4940 / 5940 student.

Student help mode: {help_mode}
Preferred language for examples: {pref_lang}

Use the context below, if relevant, to answer their question.
If the context does not answer the question, say that and answer
based on general LLM knowledge, but do not make up specific course
policies or due dates.

Context:
{context_block}

Student question:
{user_input}
"""

    # 3. Stream back to chat UI
    response_stream = await chat_client.stream_async(prompt)
    await chat.append_message_stream(response_stream)

app = App(app_ui, server)

```

## Generative AI (GAI) self-reflection

For this homework, I utilized generative AI tools for coding and help with designing reasoning prompts. I also used Generative AI to help build my shiny app. I think I demonstrated mastery of topics by effectively using batch processing to evaluate all 9 model and prompt combinations, compare them using accuracy, f1, sensitivity, and specificity. I also created visualizations to help compare model performance. Lastly, I created a shiny app using RAG to store all the homeworks,

syllabus and projects then added some extra functionality such as language/topic selections and tips/example questions.